

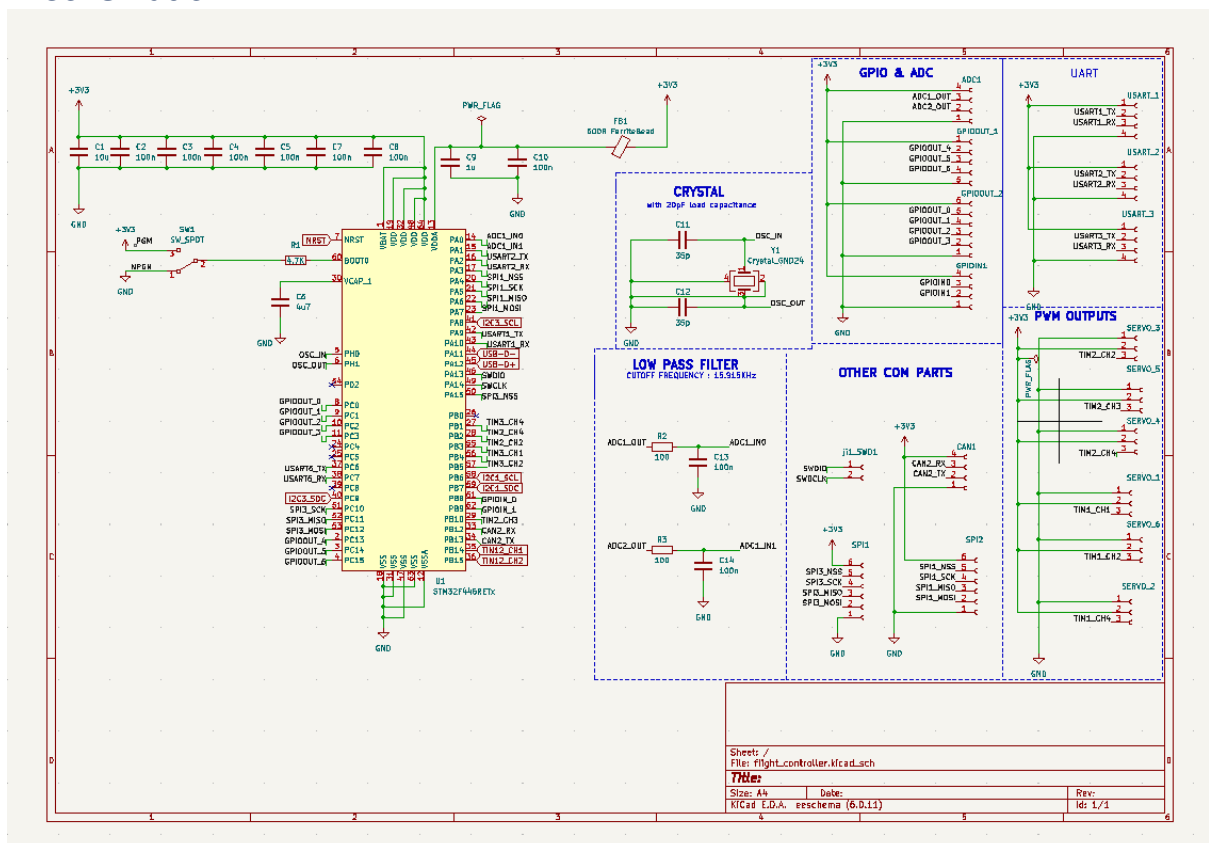
FLIGHT CONTROLLER PCB

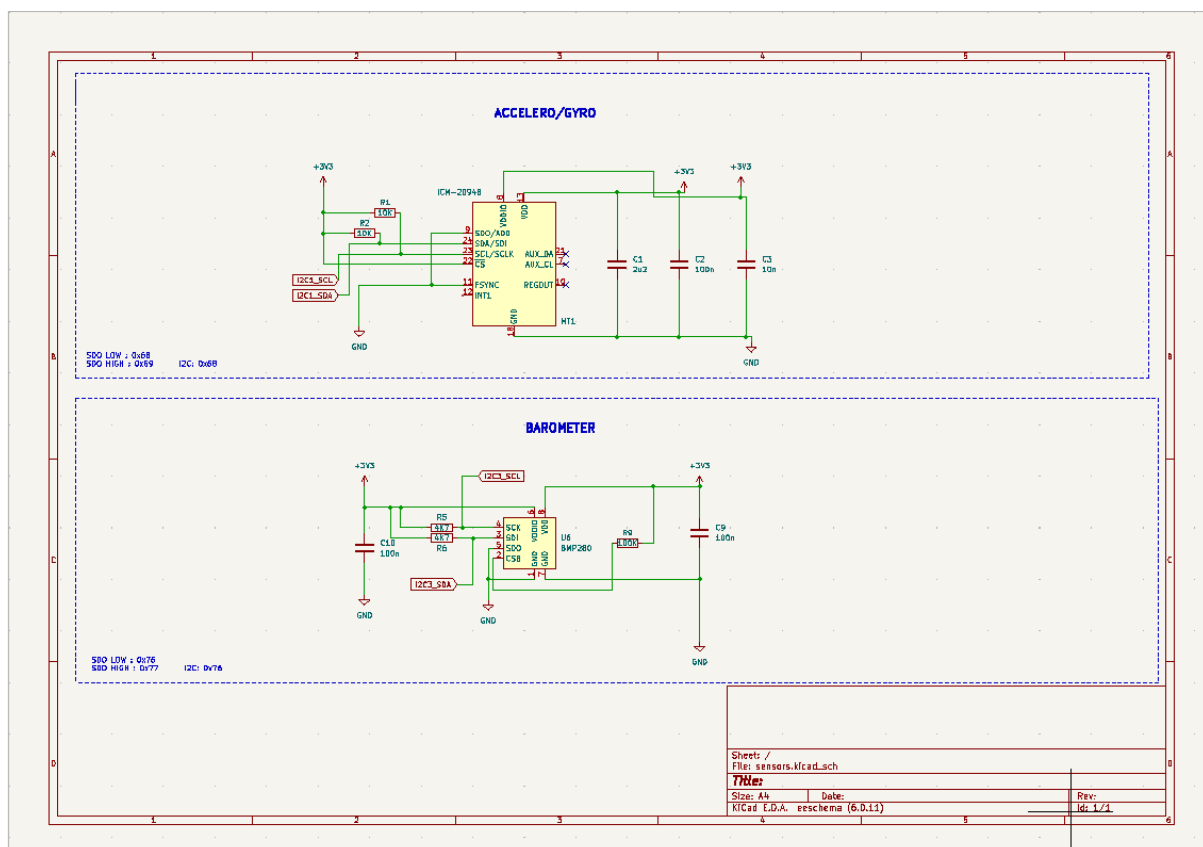
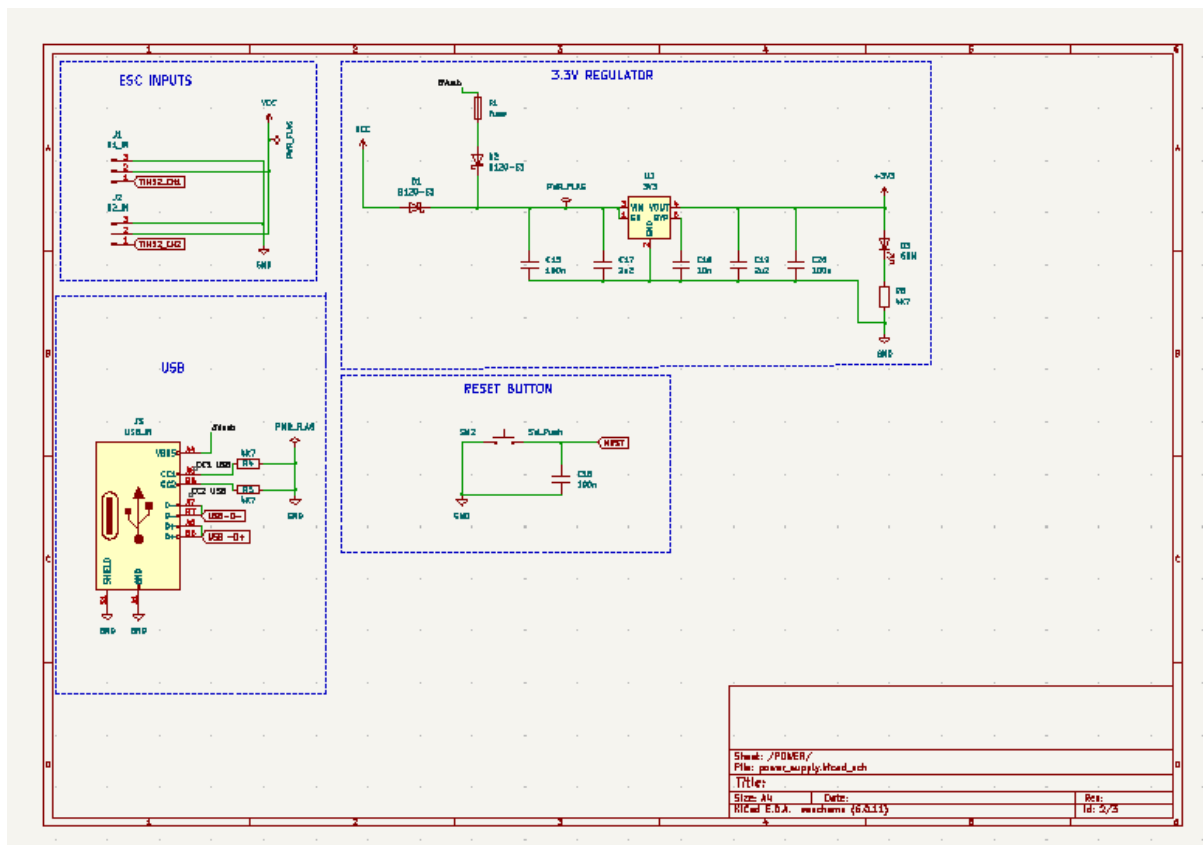
4-Layer PCB Design — STM32F446RE Based

1. Project Overview

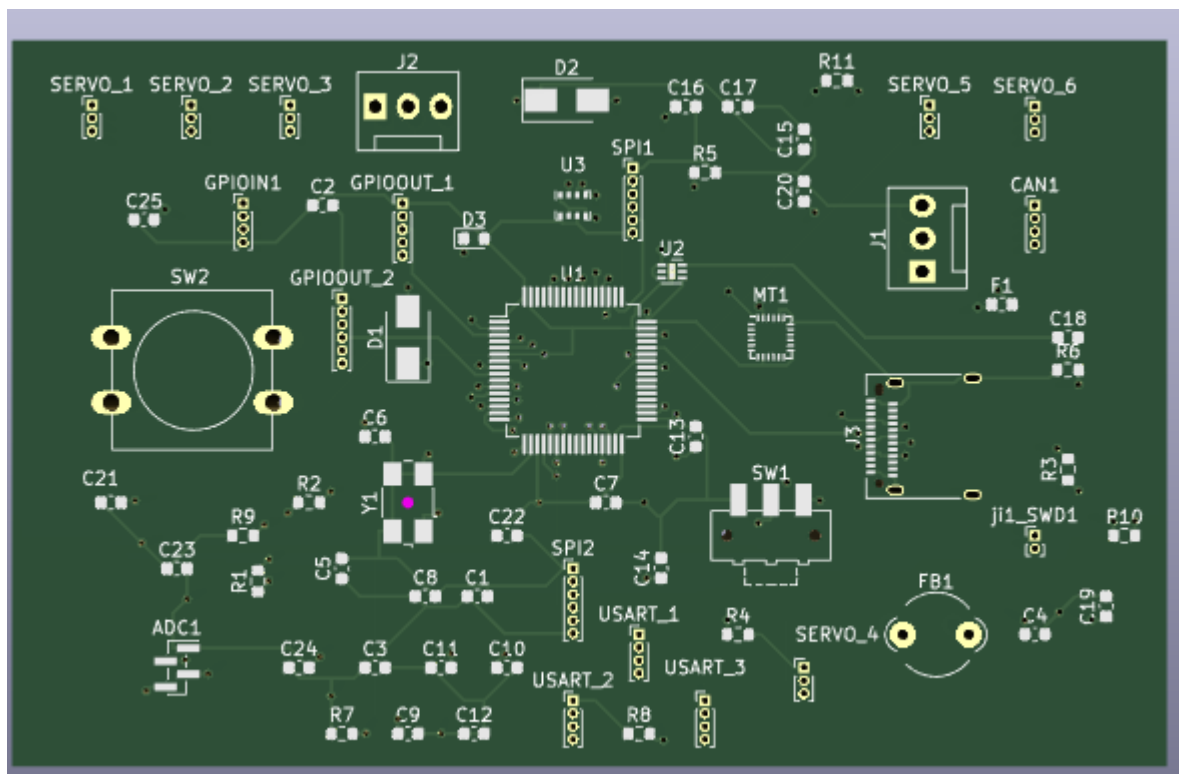
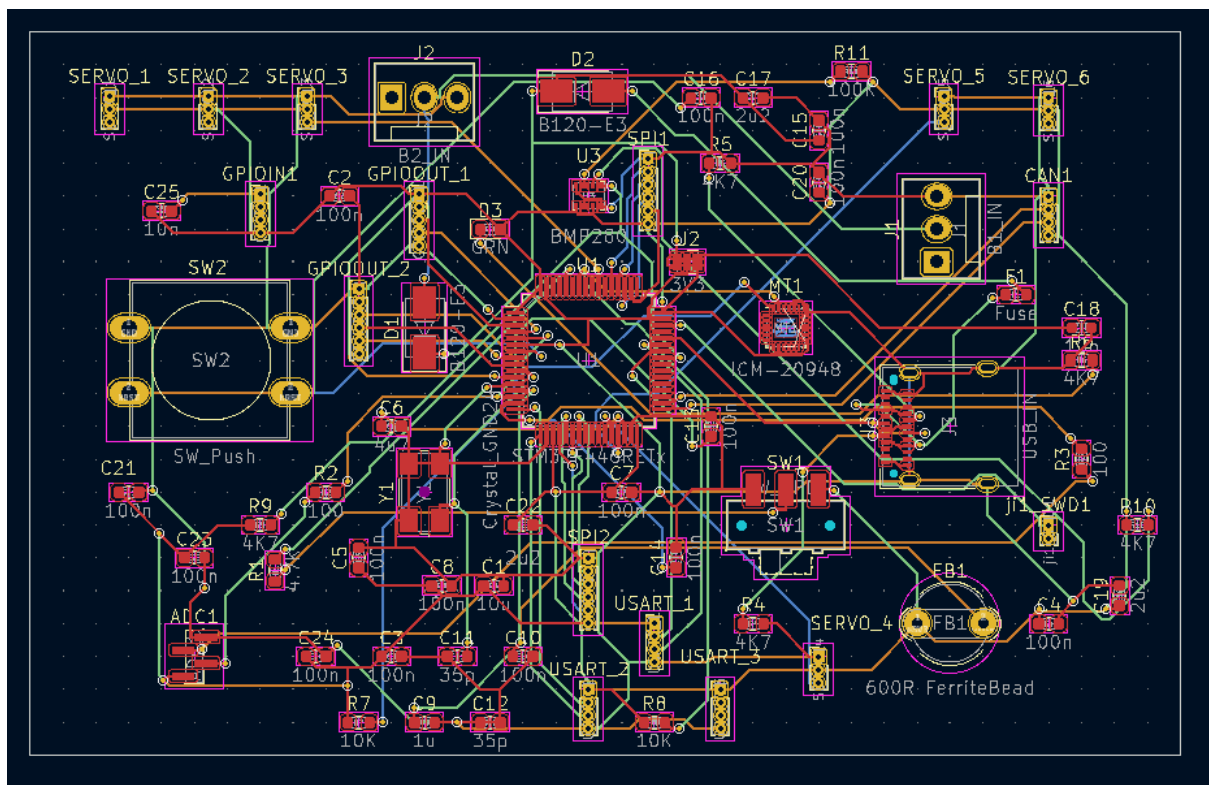
Parameter	Detail
Board Name	Flight Controller (flight_controller.kicad_sch)
Design Tool	eSim 2.5
Layer Count	4 layers
Core MCU	STM32F446RETx (ARM Cortex-M4, LQFP64)
Board Function	Flight control board with IMU/barometer sensing, 6-channel PWM servo/motor output, 3x UART, CAN, dual SPI, USB, and analog input for a UAV/drone or similar embedded control application.

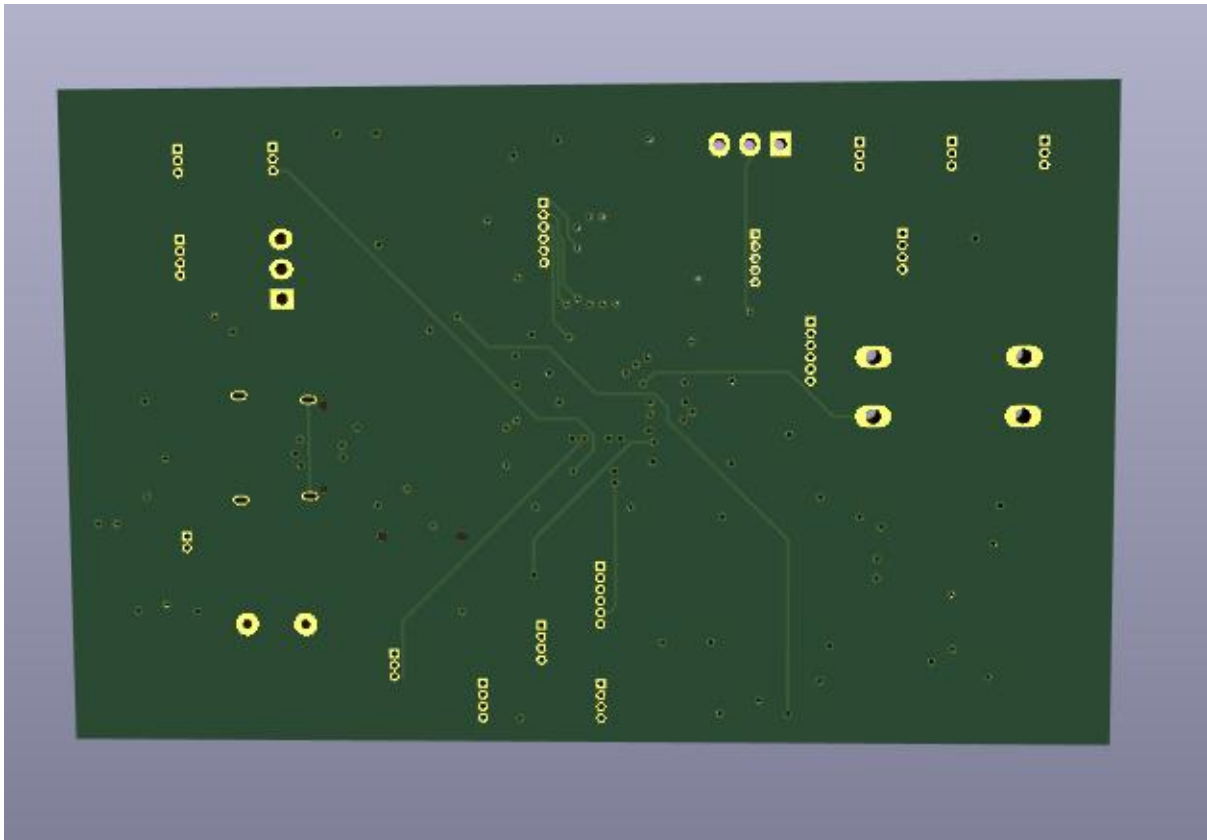
2. Schematic





3. PCB Layout & 3D Views





4. PCB Stack-Up (4-Layer)

Layer	Assignment
L1 (Top)	Signal + component placement
L2	Ground plane (GND pour)
L3	Power plane (+3V3 / PWR_FLAG pours)
L4 (Bottom)	Signal, component placement ,return routing

Notes:

- Dedicated GND and power planes are recommended (and consistent with the routing/pour density seen in the layout views) to minimize EMI, reduce loop inductance for the switching regulator, and provide a low-impedance return path for high-speed lines (USB D+/D-, SPI, CAN).

5. Core Processing Unit

Parameter	Detail
MCU	STM32F446RETx
Core	ARM Cortex-M4 (with FPU)
Package	LQFP64
Reset	NRST with pull-up (R1) and manual reset switch (SW1, SW_SPST) tied to boot/program header (labelled PGM), debounced
Boot	BOOT0 pin available via header/jumper for DFU/bootloader entry
Internal Regulator Cap	VCAP_1 decoupled with C6 = 4.7 uF (required for STM32F4 internal regulator)
Decoupling	C1-C5, C7, C8 distributed across all VDD/VSS pins: mix of 10 uF bulk & 100 nF ceramic bypass per pin pair
Clock	External crystal Y1 (HSE) with load capacitors C11 = C12 = 35 pF footprint, tuned for 20 pF load capacitance; OSC_IN/OSC_OUT routed with dedicated crystal ground guard (Crystal_GND24 footprint)
Power Filtering	Ferrite bead FB1 (600 ohm @ high freq) on PWR_FLAG rail feeding servo/PWM power domain, with bulk caps C9 (1 uF) and C10 (100 nF)

6. Power Supply Subsystem

Parameter	Detail
Input Protection	Schottky diode D1 (B120-E3) for reverse-polarity protection on VCC input
Fuse	F1 in-line fuse on input rail
Regulator	U3, 3.3 V LDO/switcher (SOT23-5, marked "XM3"), VIN/VOUT/GND/EN configuration
Input Caps	C15 = 100 nF, C17 = 2.2 uF
Output Caps	C16, C19 = 2.2 uF, C20 = 100 nF (bulk + HF bypass on +3V3 rail)
Power Indicator	LED D3 (green) with series current-limit resistor R6
Output Rail	+3V3, distributed globally to MCU, sensors, and logic-level headers
Servo Power Rail	PWR_FLAG (input voltage, e.g., battery/BEC rail) routed separately to PWM/servo headers via FB1, isolated from logic supply

7. USB Interface

Parameter	Detail
Connector	J3, USB connector with CC1/CC2 (Type-C configuration channel pins) and shield ground pins
Data Lines	USB_D+ / USB_D- routed directly to MCU (PA11/PA12 - USB_D-/USB_D+)
Power	VBUS routed to power tree (through protection/regulation)
Shield	Connector shield tied to chassis/GND separately from signal GND pour where applicable

8. Reset Circuit

Parameter	Detail
Switch	SW2, tactile push-button
Debounce	C13 = 100 nF across switch contacts
Output	Drives NRST net to MCU reset pin

9. ESC Inputs

Connector	Signal	Notes
J1	ESC1_IN (TIM2_CH1)	3-pin header: Signal, VCC, GND
J2	ESC2_IN (TIM2_CH2)	3-pin header: Signal, VCC, GND

Powered from PWR_FLAG rail (raw/battery-side voltage), independent of the 3.3 V logic rail.

10. Sensor Subsystem

10.1 IMU (Accelerometer / Gyroscope / Magnetometer)

Parameter	Detail
IC	ICM-20948 (9-axis: 3-axis accel, 3-axis gyro, 3-axis mag)
Interface	I2C1 (SDA/SCL) - SDA/SDI and SCL/SCLK pins
I2C Address	0x68 (AD0 = LOW) / 0x69 (AD0 = HIGH), selected via R1/R2 (10 kOhm) configuration
Aux I2C	AUX_DA / AUX_CL broken out (for magnetometer passthrough or additional aux sensor)
Interrupt/Sync	FSYNC and INT1 routed to MCU
Decoupling	C1 = 2.2 uF, C2 = 100 nF, C3 = 10 nF
Regulator Output Cap	REGOUT decoupled via MT1
Supply	+3V3

10.2 Barometer

Parameter	Detail
IC	BMP280
Interface	I2C3 (SCL/SDA), configurable to SPI (SCK/SDI/SDO/CSB pins present)
I2C Address	0x76 (SDO = LOW) / 0x77 (SDO = HIGH)
Pull-ups	R5 = 4.7 kOhm, R6 = 4.7 kOhm on SCL/SDA
Decoupling	C9 = 100 nF, C10 = 100 nF
Supply	+3V3

11. Communication Interfaces

Interface	Channels	Header Signals
UART	USART1, USART2, USART3	TX, RX, GND (3-pin headers each)
SPI	SPI1, SPI2	NSS, SCK, MISO, MOSI (per bus) - MCU SPI3/SPI4 peripheral pins mapped to header labels SPI1/SPI2
CAN	CAN1	CAN_RX, CAN_TX (via CAN2_RX/CAN2_TX MCU peripheral) - 3-pin header with +3V3/GND
I2C	I2C1 (IMU), I2C3 (Barometer), I2C4 pins also available on header breakout	SDA, SCL
SWD	Debug/Programming header (J1_SWDD1)	SWDIO, SWDCLK, +3V3, GND, NRST

12. PWM / Servo Outputs

Header	Timer / Channel
SERVO_1	TIM1_CH4
SERVO_2	TIM1_CH2
SERVO_3	TIM2_CH2
SERVO_4	TIM2_CH4
SERVO_5	TIM2_CH3
SERVO_6	TIM1_CH1 / TIM1_CH3 (per header block)

- 6 total 3-pin PWM headers (Signal, PWR_FLAG, GND), suitable for ESC or servo signal output.
- Powered from the filtered PWR_FLAG rail (through FB1 ferrite bead), isolated from MCU 3.3 V logic supply to prevent servo-induced switching noise from coupling into analog/logic domains.

13. GPIO, ADC & Digital I/O

Block	Detail
ADC1	ADC1_IN0, ADC1_IN1 inputs, each conditioned through an RC low-pass filter
Low-Pass Filter	$R2 = 100\ \Omega$, $R3 = 100\ \Omega$ with $C13 = 100\ \text{nF}$, $C14 = 100\ \text{nF}$ -> cutoff frequency = 15.915 kHz ($f_c = 1 / 2\pi R C$)
GPIO Outputs	GPIOOUT_1 through GPIOOUT_6, broken out to individual 3-pin headers (+3V3/Signal/GND)
GPIO Inputs	GPIODIN1, GPIODIN2 breakout headers

14. Connector Summary

Designator	Function
J1, J2	ESC signal inputs
J3	USB (Type-C style, CC1/CC2 present)
SERVO_1-6	PWM outputs

Designator	Function
USART_1/2/3	UART headers
SPI_1/2	SPI headers
CAN1	CAN bus header
ADC1	Analog input header
GPIOOUT_1-6, GPIOIN1/2	General-purpose digital I/O
J1_SWD1	SWD programming/debug header
SW1, SW2	Boot/program switch, reset switch

15. Key Bill of Materials (Critical Components)

Ref	Part	Function
U1	STM32F446RETx	Main MCU
U3	3.3 V regulator ("XM3" marked, SOT23-5)	Power regulation
U (IMU)	ICM-20948	9-axis IMU
U6	BMP280	Barometric pressure sensor
Y1	Crystal (HSE, 20 pF load)	MCU clock reference
D1	B120-E3 Schottky	Reverse-polarity protection
D3	LED (Green)	Power indicator
FB1	Ferrite bead, 600 ohm	Servo rail power filtering
F1	Fuse	Input overcurrent protection
SW1	SPST switch	Boot/program select
SW2	Tactile push button	Manual reset